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Hierarchical neurobehavioral model reveals that shared flexibility, not individual stability, supports rhythmic coordination

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ABSTRACT

Interpersonal coordination requires balancing individual control with interaction-derived synergy, yet it remains unclear when neural coupling contributes beyond behavior. Using an fNIRS hyperscanning paradigm, we examined dyadic rhythmic coordination and jointly modeled behavioral stability, dispositional structure, and interbrain synchrony within a hierarchical neurobehavioral framework. Across models, mean individual stability was negatively associated with dyadic performance, whereas interaction-derived shared flexibility (i.e., dyad-level behavioral stability synergy) was the most robust positive predictor. Incorporating dispositional structure showed that larger within-dyad differences in figure-embedding performance impaired coordination, whereas higher dyad-level self-esteem facilitated coordination. The neural coupling index showed no reliable main effect after accounting for behavioral and trait factors, but moderation analyses indicated a conditional contribution: interbrain synchrony compensated when shared flexibility was low, with diminishing benefit as synergy increased. Together, these findings support a hierarchical neurobehavioral architecture in which shared flexibility provides the primary foundation of coordination, dispositional structure shapes the conditions for synergy, and interbrain synchrony contributes in a context-dependent manner.

1. Introduction

Coordinated behavior between two individuals requires managing the tension between autonomous action and shared control. When people act together, they must maintain their own motor intentions while simultaneously forming a joint movement pattern that supports collective goals. This dual requirement has long been recognized in joint action research, where theories of interpersonal coordination emphasize that successful cooperation depends on how individuals align their internal states, anticipate one another, and stabilize a shared behavioral rhythm. These observations raise a central question: what determines whether two individuals can function as a single coordinated unit rather than as two loosely coupled performers.

At the behavioral level, two influential theoretical accounts offer contrasting perspectives on how interpersonal coordination attains stability. Leading-following models propose that coordination may be supported by asymmetric control, whereby the more stable or skilled partner provides a temporal reference that facilitates alignment for the less stable partner (Konvalinka et al., 2010; Repp and Su, 2013). These accounts imply that differences in the partners' intrinsic control tendencies may shape the flow of influence within the dyad and determine whether coordination takes on a more hierarchical form. From this perspective, collective performance could, in principle, reflect the influence of the stronger individual. In contrast, coordination dynamics theories characterize coordination as an emergent property of bidirectional coupling, in which partners continuously adapt to one another

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until a shared attractor state is formed (Haken et al., 1985; Kelso, 1995; Schmidt and Richardson, 2008). From this perspective, the formation of a stable coordinative pattern may rely on behavioral resources contributed by both partners, since the stability of the joint attractor reflects properties of the coupled system rather than either individual alone. While intrinsic stability is sometimes viewed as beneficial for controlling rhythmic actions, coordination dynamics frameworks imply that too much internal rigidity may impede the emergence of a shared attractor. These frameworks also acknowledge that intrinsic timing tendencies and individual stability characteristics provide the initial conditions under which coordination unfolds, potentially constraining or facilitating the emergence of interpersonal adaptation. Together, these perspectives highlight an open question regarding the origins of coordinated stability: whether it is primarily shaped by individual capacities or emerges from collective dynamics that form through interaction.

Because coordinated action is enacted through human effectors, the behavioral mechanisms described above are unlikely to operate independently of the partners' dispositional characteristics. The present study focuses on two trait domains that are theoretically aligned with the core demands of interpersonal coordination: cognitive style and self-related evaluation. Cognitive style, often indexed by figure-embedding performance, reflects how individuals extract structure from complex stimuli and construct stable internal task models. Prior work shows that individuals with more field-independent tendencies form more robust internal representations and demonstrate superior control in temporally constrained motor tasks (Höffler, 2010; Witkin et al., 1977). This representational stability is essential for mutual adaptation in coordination dynamics, as partners must maintain sufficiently compatible task models to converge on shared coordinative patterns. At the same time, excessively rigid internal representations may limit the flexibility required for partners to allow their action systems to become permeable to one another, a capacity that is central to the emergence of dyadic synergy. Self-esteem provides a complementary dispositional route through which interpersonal coordination may be shaped. Research indicates that self-esteem influences role tendencies during interaction, affecting initiative, responsiveness, and the willingness to assume or relinquish control (Campbell and Lavallee, 1993; Tice, 1993). These tendencies align closely with the asymmetries posited in leading–following models, suggesting that dyads with divergent self-esteem profiles may differ in how readily they adopt hierarchical versus reciprocal coordination strategies (Anderson et al., 2001). Such role-related tendencies may either support or disrupt the formation of a shared behavioral solution, depending on whether partners can balance influence without constraining mutual adaptability. Accordingly, cognitive style and self-esteem do not act as direct determinants of behavioral outcomes but instead define the potential pathways through which coordination develops. They set the dispositional conditions under which stability is more likely to arise from individual influence or from emergent synergy, positioning dyadic trait structure as a central element in the architecture of coordination. In this sense, dispositional factors help determine whether coordination is driven by individual stability or whether partners can cultivate the shared flexibility that allows a genuinely collective dynamic to form.

Beyond behavioral and dispositional factors, interpersonal coordination also unfolds at the neural level. Hyperscanning studies have consistently shown that interbrain synchrony increases when partners share attention, coordinate their actions, or engage in cooperative problem solving (Cui et al., 2012; J. Jiang et al., 2012; R. Niu et al., 2024). Such findings have led to the proposal that neural alignment supports real-time coordination by facilitating shared representations or predictive coupling across partners (Konvalinka and Roepstorff, 2012). However, the functional interpretation of interbrain synchrony remains debated. Some accounts view neural coupling as a marker of emergent dyadic synergy, reflecting the formation of a joint coordinative system (Cui et al., 2012; R. Niu et al., 2024; Pan et al., 2017), whereas others propose that it is recruited when coordination demands are high or behavioral alignment is unstable, potentially serving a compensatory function (Djalovski et al., 2021; Fishburn et al., 2018; Kurihara et al., 2022). These contrasting interpretations raise the possibility that neural alignment may not uniformly strengthen coordination but may instead become more pronounced when partners struggle to achieve sufficient behavioral flexibility for mutual adaptation. In such cases, neural synchrony could reflect increased permeability between partners' action systems, supporting coordination when individual stability or interpersonal fit constrains the emergence of a shared behavioral solution. Critically, existing research provides limited insight into the conditions under which neural coupling contributes to coordinated performance or how it interacts with the behavioral mechanisms and trait structures described above. It remains unclear whether neural synchrony reinforces mutual adaptation, compensates for reduced behavioral synergy, or emerges selectively under states of heightened coordination demand. Whether interbrain synchrony strengthens mutual adaptation, augments individual-driven stability, or emerges selectively during coordination challenges remains unclear. Addressing these questions requires a framework capable of examining neural coupling in conjunction with behavioral dynamics and dyadic dispositions, thereby clarifying how coordination is supported across multiple levels of organization.

Together, these considerations highlight the need for an integrated account of interpersonal coordination that simultaneously incorporates behavioral mechanisms, dispositional structure, and neural coupling. The present study addresses this gap by examining how dyads coordinate rhythmic actions while their brain activity is recorded simultaneously using fNIRS hyperscanning. This naturalistic yet controlled paradigm allows us to characterize the stability of each individual's motor dynamics, the collective properties that emerge at the dyadic level, and the conditions under which neural synchrony contributes to coordinated performance. To capture these multilevel processes, we constructed a set of hierarchical neurobehavioral models that separately quantify individual stability, dyadic coordination performance, and the modulatory role of trait composition. This modeling framework enables us to separately evaluate partners' overall stability, their stability asymmetry, and their interaction-derived shared flexibility (i.e., dyadic stability synergy), while also testing whether dispositional characteristics and neural alignment shape how these behavioral mechanisms operate within dyads. We further examined whether interbrain synchrony provides additional explanatory power beyond behavioral dynamics and whether its contribution depends on the dispositional and coordinative state of the dyad. Guided by this multilevel framework, we focused on three central research questions:

(1) What is the primary source of stability in dyadic motor coordination: the partners' overall level of individual stability, the stability asymmetry between them, or a form of emergent dyadic stability arising from their interaction?
(2) When does coordination instead reflect emergent dyadic synergy that cannot be reduced to the properties of either individual?
(3) Under what conditions does interbrain synchrony provide additional explanatory power for coordinated performance, and how does it interact with dispositional and behavioral states of the dyad?

Based on existing theory and empirical evidence, we formulated the following hypotheses:

H1. Individual Stability Hypothesis: Greater individual motor stability will enhance dyadic coordination performance, consistent with models emphasizing asymmetric, individual-driven control. This hypothesis allows us to test whether coordination is primarily supported by individual robustness or whether stability may instead depend on interaction-derived processes.

H2. Synergy Hypothesis: Greater compatibility in dyadic trait composition and behavioral dynamics will predict stronger emergent

coordination beyond individual contributions. This hypothesis focuses on whether coordination stability emerges from partners' ability to form a shared behavioral solution rather than from their individual attributes.

H3. Conditional Neural Coupling Hypothesis: Interbrain synchrony will contribute to coordination only under specific behavioral or dispositional conditions, such as when stability is difficult to maintain or when dyads exhibit strong synergy. This hypothesis reflects the view that neural alignment may operate as a context-dependent mechanism rather than a universal driver of coordination.

Through this framework, we aim to clarify how individual capabilities, dyadic organization, and neural coupling jointly support collective behavior across multiple levels of analysis.

2. Method

2.1. Participants

In the current study, 32 right-handed healthy volunteers were organized into 16 same gender dyads (9 male dyads and 7 female dyads, mean age 21.75 ± 2.11 years). An a priori power analysis was conducted in G*Power (3.1.9.7), assuming a medium effect size ($f = 0.25$), $\alpha = 0.05$, and 0.8 power. The analysis indicated a minimum required sample size of 14 dyads. We therefore recruited 32 participants forming 16 same-gender dyads to ensure adequate power and to account for potential exclusions. To minimize the influence of sociological factors such as gender composition and prior familiarity, each dyad consisted of two individuals of the same gender who did not know each other before the experiment. Handedness was determined based on participants' self-reported habitual hand use in daily activities, including writing and lifting objects. All participants had normal or corrected-to-normal vision and reported no history of physical disability or mental illness. Written informed consent was obtained from all participants, who received 100 RMB (approximately 15 USD) as compensation. The study adhered to the ethical principles of the Declaration of Helsinki and was approved by the institutional ethics committee of Shanghai University of Sport in China (tracking number 102772019RT007).

2.2. Task and procedures

A modified inter-individual coordination paradigm was used. Each participant sat beside his or her partner facing the testing table with the designated shoulder positioned at about 50 degrees of flexion, 0 degrees of abduction, and 90 degrees of medial rotation. The forearm rested on the table, the elbow was stabilized at about 120 degrees of flexion, and the wrist and forearm were kept in neutral positions. The designated hands (left for the participant on the left and right for the participant on the right) were placed at the center of the table under a high-speed camera positioned roughly 0.6 m above.

Participants performed a cyclic pinching movement with the thumb and index finger, opening widely and closing to make accurate fingertip contact. They were asked to coordinate their movements with their partner in symmetric or alternate modes, guided by a metronome. In the symmetric mode, both members of the dyad placed their fingers at the same point in the movement cycle. In the alternate mode, they placed their fingers at opposite points in the cycle. A research assistant demonstrated both modes, and participants practiced before the formal task.

The experiment followed a randomized block design with 2-phase modes and 6 frequencies, producing 48 blocks per dyad. Conditions were randomly arranged. Each block lasted 10 s with a 20 s rest interval.

2.3. Behavioral data recording

A digital video camera (HDR CX700E, Sony, Japan) recorded the cyclic pinching movements at 50 frames per second with a resolution of 1920 × 1080, and fluorescent markers placed on the fingertips enabled the extraction of displacement trajectories for both partners.

2.4. Hemodynamic data acquisition

A multichannel continuous wave fNIRS instrument (NIRScout; NIRx Medical Technologies LLC, Minneapolis, MN, United States) was used to record hemodynamic activity from both participants simultaneously throughout the experiment, with a sampling rate of 7.81 Hz. For each dyad, the two participants were fitted with identical head caps, and the optodes were arranged according to the international 10/20 system. Each head cap contained three optode probe sets, consisting of seven emitters and seven detectors with 3 cm source detector separation, forming fifteen measurement channels positioned over the premotor and bilateral inferior parietal regions (see Fig. 1). Based on the optode montage aligned to the international 10/05 system and standard scalp-to-cortex mapping, all fNIRS channels were assigned approximate anatomical labels and MNI coordinates. Channel locations, anatomical labels, and corresponding MNI coordinates are provided in Supplementary Material 1.

2.5. Measures

2.5.1. Embedded figures test

Field dependence-independence was assessed using the Chinese version of the Group Embedded Figures Test (GEFT). The original GEFT was developed by Witkin and colleagues and has been widely used with adolescent and adult samples, showing good reliability and validity as a measure of cognitive style (Witkin et al., 1971). The test consists of simple figures embedded within more complex patterns. Participants are instructed to locate and trace each simple target figure within the complex figure as accurately as possible, keeping the length and orientation of the lines consistent with the model under a specified time limit. The test includes one practice part and two formal parts. The practice items familiarize participants with the procedure and are not scored. Each formal part contains 10 items and has a time limit of 6 min. According to the test manual, items are weighted 0.5, 1, or 1.5 points depending on difficulty. A response is counted as correct when the traced figure matches the sample in both line length and orientation. The sum of the scores from the two formal parts yields the total GEFT score, with higher scores indicating a more field-independent cognitive style. In the present study, we used the sample mean of the total score as the cut-off, classifying participants scoring above the mean as field independent and those scoring at or below the mean as field dependent.

2.5.2. Rosenberg self-esteem scale

Global self-esteem was measured with the Chinese version of the Rosenberg Self-Esteem Scale (RSES). The original RSES consists of 10 items rated on a 4-point Likert scale ranging from 1 (strongly disagree) to 4 (strongly agree), with approximately half of the items worded in the negative direction and therefore reverse scored (Rosenberg, 1965). The RSES is one of the most widely used instruments for assessing general self-worth and self-acceptance, and numerous studies have demonstrated its good internal consistency, factorial validity, and criterion-related validity across diverse cultural and age groups (Gray-Little et al., 1997; Huang and Dong, 2012; Schmitt and Allik, 2005). Studies with Chinese samples further support the reliability, unidimensional factor structure, and measurement invariance of the Chinese RSES among children, adolescents, and university students (C. Jiang et al., 2023; Leung and Wong, 2008; Peng et al., 2024; Wu et al., 2017). In the present study, participants rated all 10 items; reverse-keyed items were recoded, and the item scores were summed to obtain a total self-esteem score, with higher scores indicating higher global self-esteem. The RSES required approximately 2 min to administer and, together with the GEFT, constituted the questionnaire component of the study.

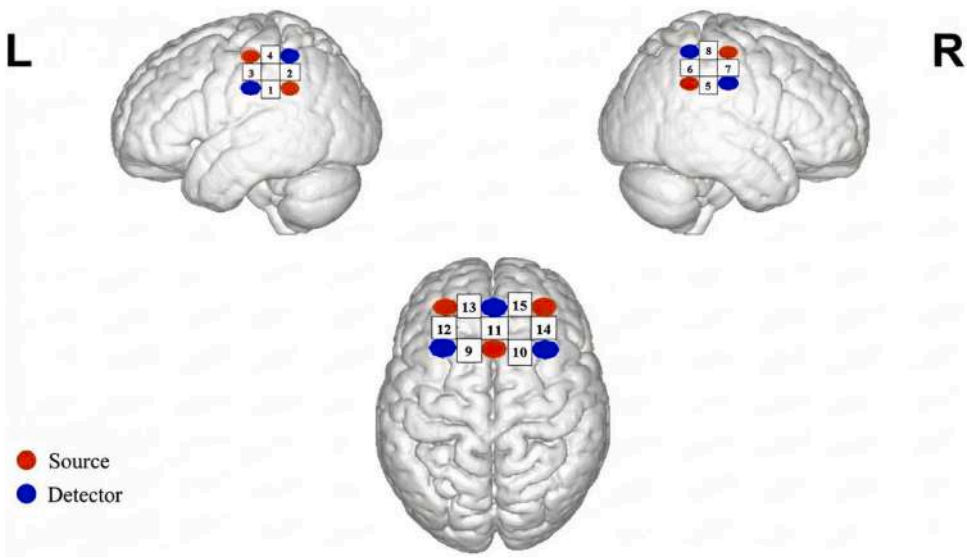

Fig. 1. Optode arrangement and channel layout for hyperscanning acquisition. Both participants in each dyad were fitted with identical fNIRS head caps containing seven light sources (red) and seven detectors (blue), forming fifteen channels over the premotor and bilateral inferior parietal regions. Lateral and superior views illustrate the symmetrical optode configuration used for simultaneous dual-brain recording.

2.6. Data analysis

2.6.1. Behavior performance

The continuous trajectories were segmented into 12 task conditions, each comprising four blocks. For each block, a 10 s displacement time series was extracted for each participant and transformed using the Hilbert method to obtain the instantaneous phase $\phi(t)$. Phase continuity was ensured by unwrapping the analytic phase. Individual rhythmic stability was quantified as the within-block standard deviation of the first-order phase difference $\Delta\phi(t)$(Eq. (1)). This stability index was computed separately for each partner, yielding Stability A (SA) and Stability B (SB), which served as the basis for subsequent dyad-level behavioral metrics. Dyadic coordination was quantified using the phase-locking value (PLV), computed from the instantaneous dyadic phase difference $\Delta\phi_{\mathrm{dyad}}(t)$(Eq. (2)).

$$\Delta\varphi(t) = \varphi(t+1) - \varphi(t), \quad \sigma^{ind}_{\Delta\varphi} = SD_t(\Delta\varphi(t)) \tag{1}$$

$$\Delta\varphi_{dyad}(t) = \varphi_A(t) - \varphi_B(t), \quad PLV = \left|\frac{1}{N}\sum_{t=1}^{N}\exp\left(i\Delta\varphi_{dyad}(t)\right)\right| \tag{2}$$

Higher values indicated stronger interpersonal phase alignment. Together, SA and SB characterize how consistently each partner maintains their intrinsic motor dynamics, whereas PLV captures the quality of the dyad's moment-to-moment coordination. This structure allows us to examine how individual-level stability properties combine to shape dyadic performance. Because dyadic coordination emerges from the interaction between two partners rather than from either member alone, subsequent analyses transformed SA and SB into dyad-level composites reflecting overall stability, stability asymmetry, and interaction-based synergy. These constructs formed the behavioral inputs of Model 1 described below. All indices were computed at the block level for subsequent statistical analyses across conditions.

2.6.2. Interbrain synchrony (IBS)

First, we conduct an initial assessment of channel quality and exclude channels whose coefficient of variation exceeds 15 percent (R. Niu et al., 2019; Piper et al., 2014). The raw optical intensity signals are then converted into optical density signals. To suppress global physiological fluctuations in the fNIRS signals, a principal component analysis-based spatial filtering technique is applied (Huppert et al., 2009; Zhang et al., 2016). Specifically, a spatial filter is constructed using the geometric information of the optodes, with the filter angle parameter set to 1.3, which regulates the degree to which spatially global components are projected out. PCA is then performed on the multi-channel signals, after which the global components identified by the spatial filter are removed before reconstructing the cleaned data. The optical density signals are subsequently transformed into hemoglobin concentration changes via the modified Beer-Lambert law. To further correct motion artifacts, a correlation-based signal improvement (CBSI) approach is applied, leveraging the negative correlation between oxygenated and deoxygenated hemoglobin changes (Cui et al., 2010). CBSI computes the ratio of the standard deviations of the two chromophores and rescales the oxygenated hemoglobin signal to restore the expected anti-correlation pattern. Given the greater sensitivity of oxygenated hemoglobin to task-related cerebral blood flow changes, all subsequent analyses focus on the HbO time series (Chen et al., 2018; Cui et al., 2012; Hoshi, 2007; H. Niu et al., 2013; R. Niu et al., 2019, 2024; Pan et al., 2017; Wang et al., 2020). All HbO time series were resampled to 1 Hz to standardize the temporal resolution for subsequent analyses. Wavelet transform coherence (WTC) was used to quantify the interbrain relationship for each dyad across all blocks. To determine the frequency band of interest, paired *t*-tests were conducted between task and resting states on a frequency-by-frequency basis, averaging across all conditions and blocks. This procedure revealed a frequency range of 8 to 10.07 s, corresponding to 0.099 to 0.125 Hz, in which WTC values during the task condition exceeded those observed during rest (see Supplementary Material 2). No correction for multiple comparisons was applied because the analysis was intended solely for identifying the frequency range exhibiting task related enhancement rather than for statistical inference (Lu et al., 2019; Zheng et al., 2018).

2.7. Analytical framework: hierarchical neurobehavioral models

In studies of social interaction, observations from two partners are inherently interdependent, which violates the independence assumption of conventional individual-level analyses. To properly account for this interdependence, we adopted a modeling strategy conceptually aligned with the Actor Partner Interdependence Model (Cook and Kenny, 2005; Kenny et al., 2006) and related dyadic analysis frameworks, which explicitly address cross-partner dependence in two-person data. This conceptual foundation was implemented using linear mixed effects

models to accommodate the repeated measures structure within dyads. At the data level, each block served as one observational unit. Every dyad completed four blocks in each of the twelve experimental conditions, yielding 48 rows per dyad. All models included a random intercept for dyad $(1|Dyad)$, to capture within dyad dependence, and fixed effects for Phase (in-phase versus anti-phase) and Rate (1.8, 2.1, 2.4, 2.7, 3.0, 3.3 Hz) to control for task structure (see Fig. 2.).

2.7.1. Model 1: behavioral stability model

Model 1 characterizes dyad-level behavioral stability by decomposing each partner's stability into three complementary components. Let *SA* and *SB* denote the raw stability indices of the two partners within a given block. To improve distributional properties and reduce skewness, the individual stability measures were first log-transformed as SA_{log} and SB_{log} (Eq. (3)). Based on these transformed scores, we defined three dyad-level stability components: mean stability S_{mean}, capturing the dyad's overall level of stability; stability difference S_{diff}, indexing the magnitude of between-partner asymmetry; and a multiplicative composite S_{dyads}, reflecting the joint magnitude of partners' stability (Eq. (4)). Dyadic performance (*DyadPerf*) was Fisher z-transformed to improve continuity and approximate normality.

$$SA_{log} = \log(SA), SB_{log} = \log(SB) \tag{3}$$

$$S_{mean} = \frac{SA_{log} + SB_{log}}{2}, S_{diff} = \left|SA_{log} - SB_{log}\right|, S_{dyads} = SA_{log} \times SB_{log} \tag{4}$$

$$DyadPerf_z = \frac{1}{2}\ln\left(\frac{1 + DyadPerf_z}{1 - DyadPerf_z}\right) \tag{5}$$

Because the experimental design manipulated both the relative phase (in-phase vs. anti-phase) and movement rate, we included the full factorial *Phase* × *Rate* term to account for systematic condition effects, capturing the main effects of *Phase* and *Rate* as well as their interaction. The full mixed-effects specification of Model 1 is summarized in (Eq. (6)). In later sections, we refer to the triplet of behavioral stability components as the Behavioral Stability construct (BS), as defined in (Eq. (7)). The model includes a random intercept for each dyad to accommodate repeated measurements across blocks and to estimate the unique contributions of the stability components while controlling for the experimental condition structure.

$$DyadPerf \sim S_{mean} + S_{diff} + S_{dyads} + Phase \times Rate + (1|Dyad) \tag{6}$$

$$BS = \left\{S_{mean}, S_{diff}, S_{dyads}\right\} \tag{7}$$

2.7.2. Model 2: behavioral and dispositional integration model

Model 2 extends Model 1 by incorporating dyad-level trait configuration. Prior work suggests that both absolute trait levels and similarity in relevant traits can shape cooperative outcomes, motivating the inclusion of trait structure at the dyadic level. In the present study, we focused on two trait domains theoretically linked to coordination, figure-embedding performance (cognitive style) and self-esteem (self-related evaluation). For a given dyad, let $Trait_A$ and $Trait_B$ denote the two partners' scores on a given scale. For each trait domain, we defined three dyad-level quantities: the dyad-level mean trait $Trait_{mean}$, indexing the overall level of the trait within the dyad (Eq. (8)); the trait difference $Trait_{diff}$, capturing between-partner asymmetry (Eq. (9)); and a dyadic trait compatibility component $Trait_{dyads}$, operationalizing similarity via an exponential decay function of the absolute difference (Eq. (10)).

$$Trait_{mean} = \frac{Trait_A + Trait_B}{2} \tag{8}$$

$$Trait_{diff} = |Trait_A - Trait_B| \tag{9}$$

$$Trait_{dyads} = \exp(-|Trait_A - Trait_B|) \tag{10}$$

These definitions were applied separately to the figure-embedding scores (*FigA, FigB*) and the self-esteem scores (*SelfA, SelfB*), yielding six dyad-level trait variables: Fig_{mean}, Fig_{diff}, Fig_{dyads}, $Self_{mean}$, $Self_{diff}$, and $Self_{dyads}$. For compactness, we refer to this set as the Trait Profile construct (TP), defined in (Eq. (11)). The mixed-effects formulation of Model 2 is summarized in (Eq. (12)). By comparing Model 2 with Model 1, we assess whether dyadic trait configuration explains additional variance in dyadic performance beyond behavioral stability, and we distinguish the contributions of overall trait level, trait asymmetry, and trait similarity.

$$TP = \left\{Fig_{mean}, Fig_{diff}, Fig_{dyads}, Self_{mean}, Self_{diff}, Self_{dyads}\right\} \tag{11}$$

$$DyadPerf_z \sim BS + TP + Phase \times Rate + (1|Dyad) \tag{12}$$

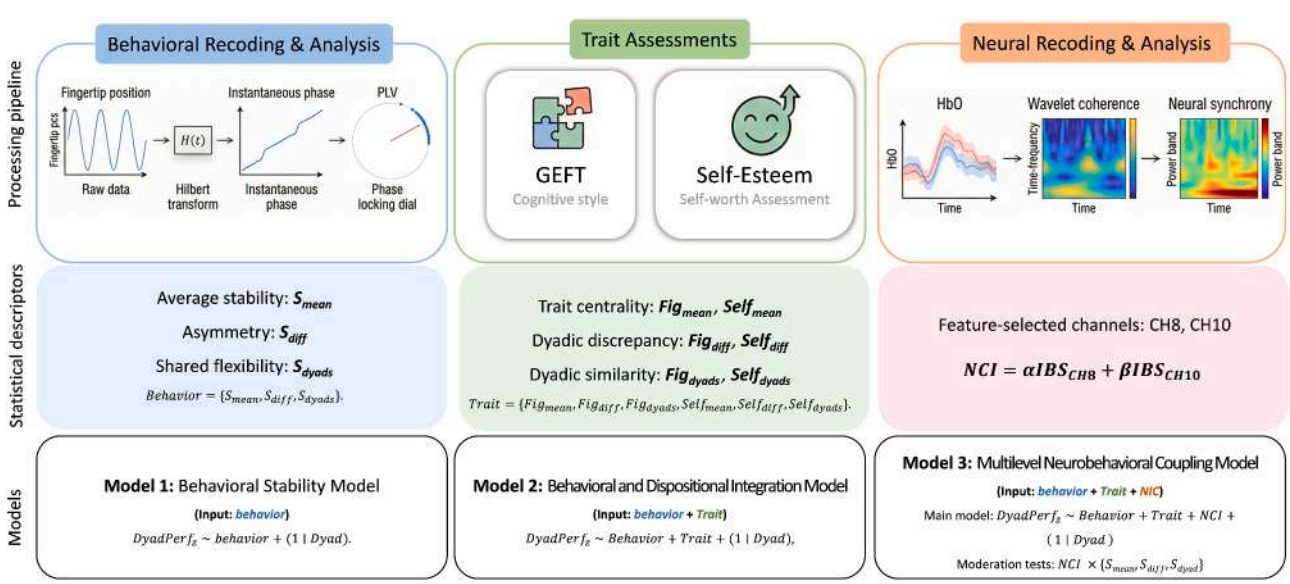

Fig. 2. Overview of the analytical framework. Behavioral, trait, and neural data were processed to derive stability descriptors, dispositional indices, and the neural coupling index. These variables were entered into three sequential models that assessed behavioral stability, behavioral trait integration, and neurobehavioral coupling in predicting dyadic performance.

2.7.3. Model 3: multilevel neurobehavioral coupling model

Model 3 incorporates a dyad-level neural coupling index derived from inter-brain synchrony and examines whether neural coupling has an independent effect on coordination performance and whether it alters the influence of behavioral stability. Hyperscanning research has demonstrated that neural synchrony between brains is a reliable marker of real-time social interaction and collaborative success across EEG, fNIRS, and other modalities, which motivates combining behavioral and neural levels of analysis.

2.8. Construction of the weighted neural coupling index

Model 3 extends the behavioral and dispositional integration model by incorporating neural coupling indices derived from inter-brain synchrony. This model aims to evaluate whether dyad-level neural coupling contributes additional explanatory power beyond behavioral stability and dispositional traits, and whether neural coupling alters the influence of behavioral stability on coordination performance. Neural information was operationalized through a weighted neural coupling index constructed from two task-sensitive cross-brain channels.

2.9. Model 3A main effect of neural coupling

To examine whether neural coupling provides an independent contribution to dyadic coordination, we extended the model by incorporating a neural coupling index (NCI) in addition to the behavioral stability construct (BS) and the dispositional trait profile (TP). Neural information was derived from inter-brain synchrony estimates across all cross-brain channel pairs. Because neural synchrony varies in its task sensitivity across channels, channel selection and weighting were determined through a data-driven procedure described in the Results section. This ensured that the neural predictor was constructed solely from task-relevant synchrony components without imposing a priori assumptions about specific channels. A weighted neural coupling index was then computed by estimating the relative contribution of task-sensitive synchrony components through a mixed effects regression using dyadic performance as the outcome. The resulting composite NCI served as the neural level predictor in the full main effect model.

2.10. Model 3B neural coupling moderation models

Rather than estimating all neural and behavioral effects within a single model, Model 3B was structured as a set of targeted moderation analyses to address a focused question: whether interbrain neural coupling modulates specific behavioral foundations of dyadic coordination. Neuroscience accounts of joint action propose that neural synchrony supports interpersonal coordination, particularly when behavioral alignment is weak, suggesting a compensatory or amplifying role of neural coupling. To test this hypothesis, Model 3B introduced interaction terms between the neural coupling index (NCI) and each behavioral stability component. Importantly, each moderation model included only one NCI × stability interaction term, while the remaining predictors were treated as covariates. All moderation models retained the full factorial structure of the task manipulations (Phase, Rate, and Phase × Rate), ensuring that neural effects were estimated independently of experimental difficulty. Accordingly, three parallel moderation models were estimated, each testing whether NCI moderates a single behavioral stability component (mean stability, stability discrepancy, or dyadic stability synergy), respectively (Eqs. (13)–15). By estimating these interaction terms, Model 3B evaluates whether neural coupling preferentially enhances performance when dyadic behavioral stability is low, or becomes redundant when the dyad already exhibits strong shared flexibility (i.e., dyad-level stability synergy), thereby characterizing cross-level dependencies between behavioral stability and neural coupling in dyadic coordinationFigure 2

$$DyadPerf \sim BS + TP + NCI + S_{mean} \times NCI + Phase \times Rate + (1|Dyad) \quad (13)$$

$$DyadPerf \sim BS + TP + NCI + S_{diff} \times NCI + Phase \times Rate + (1|Dyad) \quad (14)$$

$$DyadPerf \sim BS + TP + NCI + S_{dyads} \times NCI + Phase \times Rate + (1|Dyad) \quad (15)$$

3. Results

3.1. Model 1: behavioral stability model

Model 1 examined whether behavioral stability predicts dyadic phase coordination. The model included three stability components, S_{mean} (overall stability), S_{diff} (stability discrepancy), and S_{dyads} (stability synergy), while controlling for *Phase, Rate*, their interaction, and a random intercept for each dyad. Model fit was adequate ($AIC = 145.780$, $BIC = 187.580$). S_{mean} significantly negatively predicted dyadic performance ($\beta = -0.191$, $SE = 0.047$, $t = -4.021$, $p < .001$), indicating that higher overall individual stability was associated with poorer dyadic coordination. S_{diff} was not significant ($\beta = 0.014$, $SE = 0.012$, $t = 1.136$, $p = .256$). In contrast, S_{dyads} positively predicted dyadic performance ($\beta = 0.085$, $SE = 0.038$, $t = 2.218$, $p = .027$), suggesting that dyads with jointly high stability show better coordination. *Phase* ($\beta = -0.869$, $SE = 0.042$, $t = -20.627$, $p < .001$), *Rate* ($\beta = -0.020$, $SE = 0.008$, $t = -2.452$, $p = .016$), and their interaction ($\beta = 0.058$, $SE = 0.011$, $t = 5.397$, $p < .001$) were all significant. Together, the findings indicate that shared flexibility (synergistic behavioral stability), rather than individual stability alone, is the key behavioral predictor of dyadic coordination (see Fig. 3).

3.2. Model 2: behavioral and dispositional integration model

Model 2 added dyadic trait structures, including figure-embedding performance (*FigA, FigB*) and self-esteem (*SelfA, SelfB*), on top of the behavioral stability components. Using the same decomposition approach as in Model 1, we computed Fig_{mean}, Fig_{diff}, Fig_{dyads}, and $Self_{mean}$, $Self_{diff}$, $Self_{dyads}$. Model fit was ($AIC = 157.380$, $BIC = 227.040$). Among the behavioral predictors, S_{mean} remained significant ($\beta = -0.126$, $SE = 0.036$, $t = -3.594$, $p < .001$), and S_{dyads} remained positively associated with dyadic performance ($\beta = 0.050$, $SE = 0.025$, $t = 1.998$, $p = .046$), confirming the robustness of the behavioral stability structure. Regarding trait factors, neither Fig_{mean} ($\beta = 0.019$, $SE = 0.033$, $t = 0.596$, $p = .551$) nor Fig_{dyads} ($\beta = 0.014$, $SE = 0.041$, $t = 0.341$, $p = .733$) showed significant associations, indicating that overall cognitive ability or cognitive synergy did not predict phase coordination. In contrast, Fig_{diff} negatively predicted dyadic performance ($\beta = -0.085$, $SE = 0.042$, $t = -2.051$, $p = .040$), suggesting that larger within-dyad differences in cognitive ability impair coordination. For self-esteem, $Self_{diff}$ ($\beta < 0.001$, $SE = 0.054$, $t = 0.008$, $p = .994$) and $Self_{dyads}$ ($\beta = -0.040$, $SE = 0.051$, $t = -0.784$, $p = .434$) were not significant. However, $Self_{mean}$ positively predicted dyadic performance ($\beta = 0.066$, $SE = 0.033$, $t = 1.997$, $p = .046$), implying that dyads with higher average self-esteem levels may coordinate better, potentially because greater collective confidence and social efficacy facilitate smoother mutual adjustment and reduce hesitation during online error correction. The effects of phase, rate, and their interaction were consistent with those observed in Model 1 and are not discussed further. Overall, Model 2 demonstrates that behavioral stability remains the dominant predictor of dyadic coordination, while trait-level predictors show asymmetric effects: cognitive differences hinder coordination, whereas higher dyad-level self-esteem facilitates it (see Fig. 4).

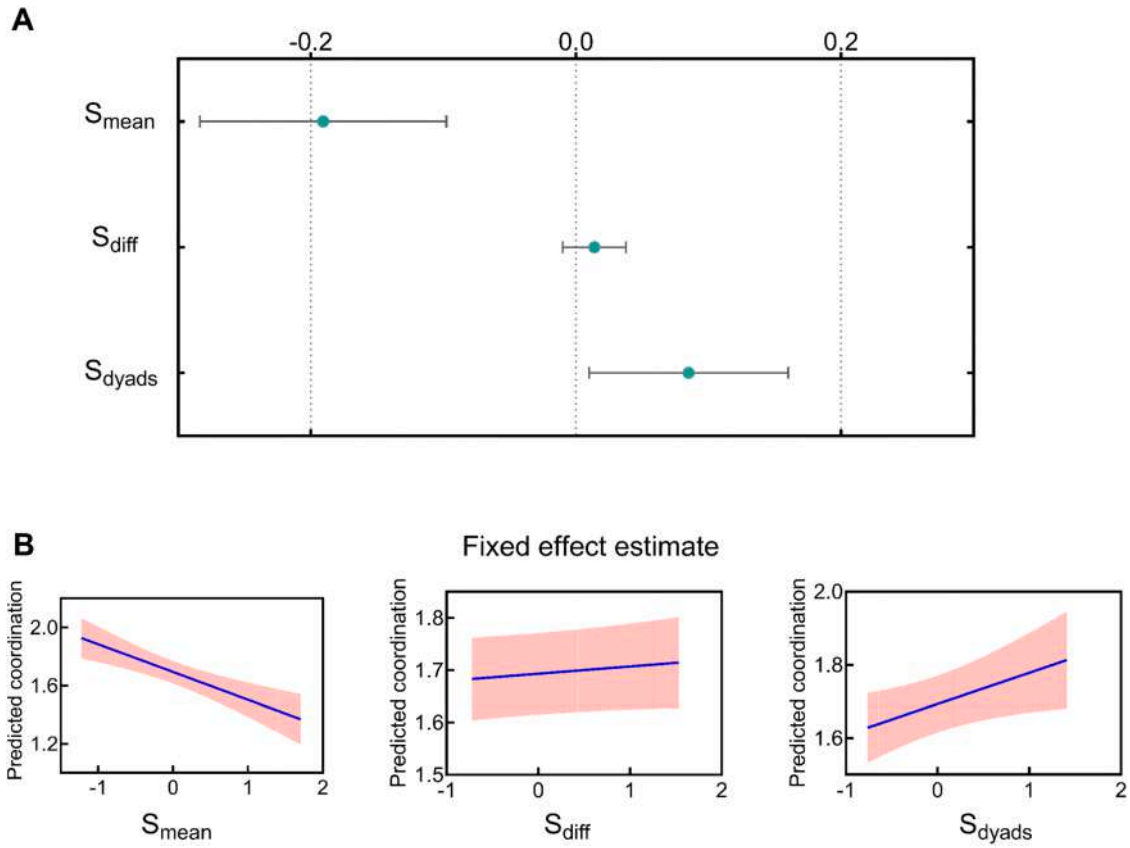

Fig. 3. Behavioral Stability Model results. Panel A shows the fixed effect estimates with standard error bars for all predictors. Panel B displays the predicted coordination as a function of each stability component. S_{mean} shows a negative association, S_{diff} shows a minimal association, and S_{dyads} shows a positive association, illustrating the opposing roles of overall and synergistic stability in dyadic coordination.

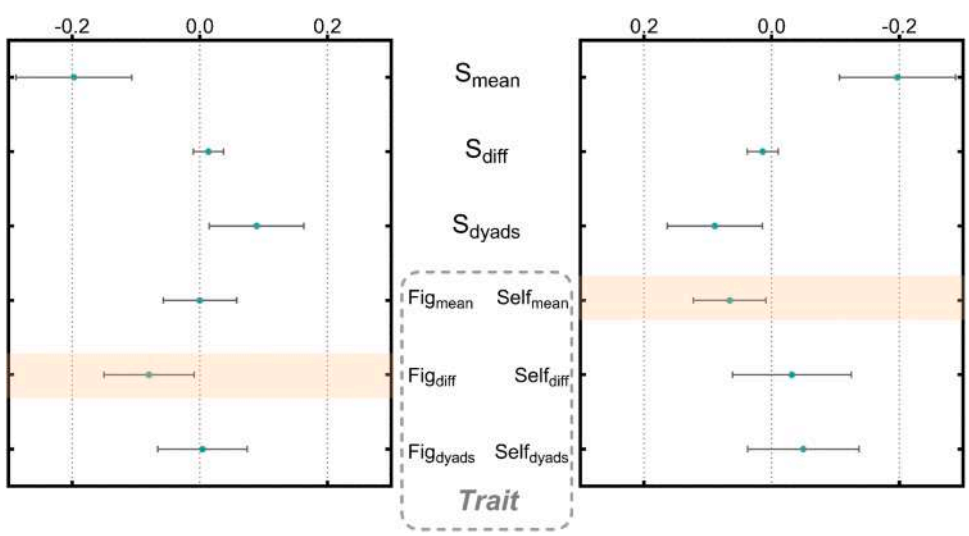

Fig. 4. Behavioral and dispositional integration model results. Panel A shows the standardized fixed effect estimates for all behavioral and trait predictors. Panel B displays the predicted coordination as a function of each trait component. Cognitive difference (Fig_{diff}) negatively predicted coordination, whereas dyadic self-esteem level ($Self_{mean}$) showed a positive association, highlighting asymmetric trait contributions to dyadic performance.

3.3. Model 3: multilevel neurobehavioral coupling model

Before constructing Model 3, we conducted paired-sample t tests comparing task and rest across all channels to identify task-sensitive interbrain synchrony. Significant effects were observed at CH8 (t = 2.064, p = .048) and CH10 (t = 3.059, p = .005), and these channels were therefore selected for subsequent analyses. According to the scalp-to-cortex mapping (see Supplementary Material 1), CH8 was localized over the right inferior parietal lobe, whereas CH10 was localized over the premotor cortex. Model 3A then tested whether the neural coupling index (NCI) explains additional variance in dyadic coordination beyond the behavioral stability and trait composition components included in Model 2. To construct NCI, we first fit a preliminary neural-weight model entering channel-level synchrony at CH8 and CH10 as separate

predictors while controlling for phase and rate. Although neither channel alone significantly predicted coordination (IBS8: $\omega 8 = -0.034$, $SE = 0.055$, $t = -0.610$, $p = .542$; IBS10: $\omega 10 = 0.057$, $SE = 0.052$, $t = 1.092$, $p = .275$), their regression coefficients were used as weights to form the composite NCI for subsequent models (see Fig. 5).

After controlling for S_{mean}, S_{diff}, S_{dyads}, figure-embedding indices, and self-esteem components, NCI did not significantly predict dyadic performance, $\beta = 1.059$, $SE = 0.825$, $t = 1.283$, $p = .200$. The behavioral stability components remained consistent with previous models: S_{mean} continued to negatively predict dyadic coordination, $\beta = -0.127$, $SE = 0.036$, $t = -3.543$, $p < .001$, and S_{dyads} remained positively associated with coordination, $\beta = 0.048$, $SE = 0.025$, $t = 1.936$, $p = .053$. Stability discrepancy (S_{diff}) again showed no significant effect, $\beta = 0.018$, $SE = 0.016$, $t = 1.142$, $p = .254$. For the trait-level components, differences in figure-embedding ability (Fig_{diff}) negatively predicted dyadic coordination, $\beta = -0.086$, $SE = 0.041$, $t = -2.055$, $p = 0.040$, while figure-embedding similarity (Fig_{dyads}) was not significant, $\beta = 0.014$, $SE = 0.041$, $t = 0.359$, $p = .720$. For self-esteem, neither self-esteem similarity ($Self_{dyads}$: $\beta = -0.038$, $SE = 0.051$, $t = -0.755$, $p = .450$) nor self-esteem discrepancy ($Self_{diff}$: $\beta < 0.001$, $SE = 0.053$, $t = 0.011$, $p = .991$) significantly predicted coordination. However, the overall level ($Self_{mean}$) showed a positive effect, $\beta = 0.067$, $SE = 0.033$, $t = 2.027$, $p = .043$, indicating that dyads composed of partners with higher aggregate self-esteem show better coordination, possibly reflecting greater confidence or reduced interpersonal hesitation during joint action. Overall, model 3A indicates that neural coupling, when added as an additional fixed effect, does not independently predict dyadic coordination once behavioral and trait components are accounted for. The dominance of the behavioral stability structure remains clear, with synergistic stability (S_{dyads}) emerging as the most consistent positive predictor across models. Besides, as a robustness check, we compared all non-empty combinations of BS, TP, and NCI (see supplementary material 3). The BS+TP model showed the best fit, with the BS+TP+NCI model performing comparably ($\Delta AIC < 1$), indicating that neural coupling provides limited improvement as a simple main effect.

Given the limited improvement when adding NCI as a uniform additive term, we next examined whether neural coupling contributes in a conditional manner by moderating the behavioral stability components. These moderation tests revealed a selective pattern, with NCI moderating overall stability and dyadic stability synergy, but not stability discrepancy. Accordingly, we estimated three parallel models, each including one NCI × stability interaction term to test whether neural coupling conditions the effect of that stability component on coordination. The interaction between S_{mean} and NCI was significant, $\beta = -2.123$, $SE = 0.996$, $t = -2.131$, $p = .033$, indicating a significant effect, suggesting that neural coupling intensifies the negative relationship between overall behavioral stability and dyadic coordination. The interaction between S_{diff} and NCI was not significant, $\beta = -1.333$, $SE = 0.924$, $t = -1.443$, $p = .149$, implying that neural coupling does not meaningfully alter the role of stability discrepancy. In contrast, the interaction between S_{dyads} and NCI was significant and negative, $\beta = -3.581$, $SE = 1.341$, $t = -2.669$, $p = .008$, revealing a compensatory moderation pattern. Simple-slope interpretation indicates that when shared flexibility (i.e., dyadic stability synergy) is low, higher neural coupling is associated with better dyadic coordination. However, when synergy is already high, the marginal benefit of neural coupling diminishes and may even reverse. This pattern suggests that neural synchrony primarily facilitates coordination when behavioral coupling is weak, but contributes less when strong shared flexibility is already present. Overall, Model 3B demonstrates that neural coupling does not uniformly amplify behavioral stability effects. Instead, it selectively compensates for low shared flexibility, revealing a conditional role of neural synchrony in shaping dyadic coordination (see Fig. 6).

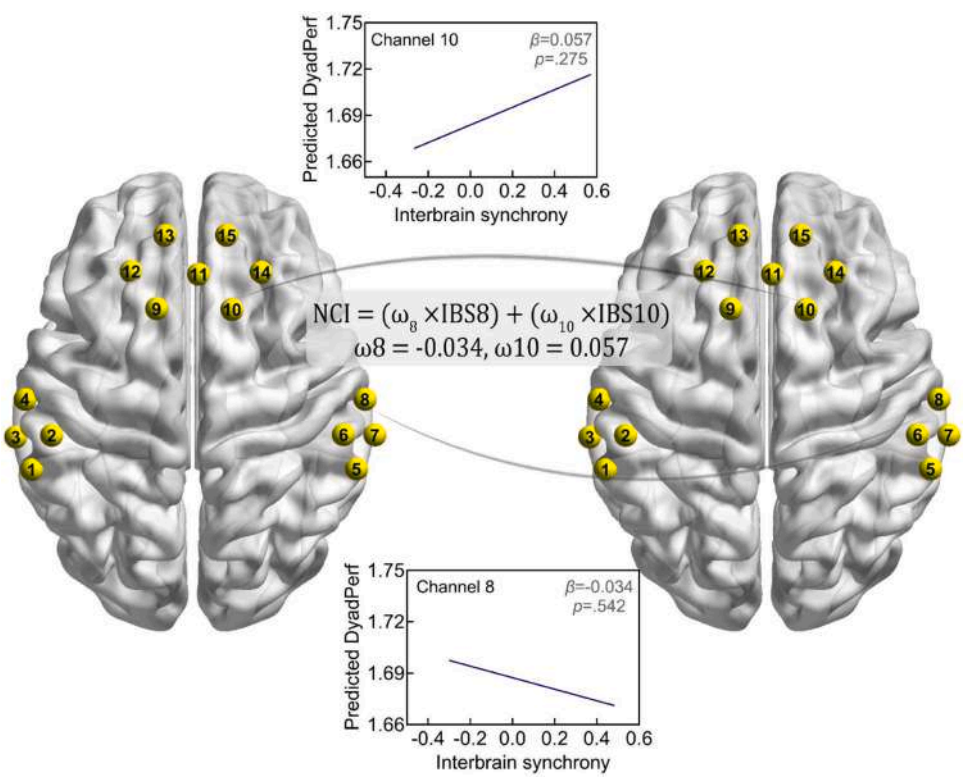

Fig. 5. Construction of the neural coupling index (NCI) from task-sensitive interbrain synchrony. Channels CH8 and CH10 were identified as task-sensitive via paired-sample *t* tests comparing task versus rest. The two brains are shown with homologous channels connected, and insets depict the fitted association between interbrain synchrony (IBS) and predicted dyadic performance from the neural-weight model (controlling for Phase × Rate), with regression coefficients (ω) and *p* values annotated. The NCI was computed as a weighted composite of the two channels, $NCI = (\omega 8 \times \text{IBS8}) + (\omega 10 \times \text{IBS10})$, using the estimated weights from the neural-weight model.

4. Discussion

The present study examined how behavioral stability, dispositional structure, and neural coupling jointly shape interpersonal motor coordination, allowing us to directly evaluate three competing hypotheses concerning individual stability (H1), interaction-derived synergy (H2), and conditional neural coupling (H3). By integrating multilevel measurements within a single dyadic framework, our findings reveal that coordinated performance does not arise from individual control capacity alone, contrary to H1, but instead reflects the dynamic balance between shared behavioral flexibility consistent with H2, interpersonal compatibility, and context-dependent neural engagement as predicted by H3. Across three hierarchical models, we identified a consistent pattern indicating that dyadic stability is best understood as an emergent property of the collective system rather than a direct function of either partner's intrinsic attributes. Together, these findings provide a coherent evaluation of the proposed hypotheses and motivate the multilevel interpretation developed below. These results highlight the importance of examining coordination through a multilayer lens, in which behavioral, dispositional, and neural processes interact to support or constrain the formation of stable joint action. The following sections elaborate on how each level of analysis contributes to this integrated account.

Our findings offer a refined perspective on how stability arises in interpersonal coordination. Rather than indicating that one partner's intrinsic stability directly supports dyadic performance, the results show that higher individual stability was associated with poorer coordination, whereas synergistic stability formed through the interaction of both partners predicted better performance. This pattern suggests that coordinated behavior depends less on unilateral control and more on both partners becoming sufficiently flexible to adapt to one another. The dyadic system achieved its most stable form when each partner softened rigid control tendencies, indicating that coordination relies on a shared flexibility that enables mutual adjustment. This interpretation is compatible with accounts that emphasize bidirectional adaptation and the emergence of shared attractor states in joint action (Haken et al., 1985; Kelso, 1995; Schmidt and Richardson, 2008). From this view, stability arises not because one partner dominates the temporal structure but because both partners permit their internal dynamics to become permeable to external influence. Our results provide behavioral evidence for this distributed control process, showing that dyadic stability is an emergent property of the collective system rather than a simple reflection of individual capacities. Such insights may inform applied contexts in which stable cooperation is required, for example, training environments that emphasize flexible mutual adjustment rather than rigidly enhancing individual precision.

Model 2 demonstrates that shared flexibility does not emerge uniformly across dyads but is shaped by partners' dispositional characteristics. Larger discrepancies in cognitive style impaired coordination, consistent with evidence that shared representational frameworks facilitate interpersonal prediction and joint control (Witkin et al., 1977; Höffler, 2010). When partners differed substantially in figure embedding performance, their internal task models were likely misaligned, limiting their ability to relax individual control and converge on a shared behavioral solution. In contrast, $\text{Self}_{\text{mean}}$ positively predicted dyadic coordination, highlighting that higher shared levels of self-worth may help partners navigate interpersonal demands more effectively. Rather than differences in self-esteem, it was the dyads' overall self-esteem level that supported coordination, aligning with evidence that self-related security facilitates openness, willingness to yield, and reduced threat sensitivity during interaction (Campbell and Lavallee, 1993; Tice, 1993; Anderson et al., 2001). Asymmetric self-related dispositions may facilitate complementary social roles, reducing competition for control and allowing partners to distribute influence more efficiently when forming a collective system. Importantly, cognitive style and self-related evaluation reflect distinct psychological domains, one concerning how individuals construct internal task models and the other concerning how they negotiate social roles. Whereas larger discrepancies in cognitive ability impaired coordination, higher dyad-level self-esteem facilitated it, indicating that these domains shape collective behavior through independent processes rather than reflecting a shared underlying disposition. Together, these findings indicate that shared flexibility is not a passive reduction of control but a regulated interpersonal state that requires both cognitive compatibility and social role differentiation. Dyads must be sufficiently aligned in how they represent the task yet sufficiently grounded in collective confidence to avoid control conflicts and sustain joint stability. Under these conditions, both partners become able to soften their stability constraints and allow a new dyadic attractor state to emerge.

Model 3 provides critical insight into the neural basis of the behavioral effects described above. Although hyperscanning studies often interpret interbrain synchrony as a driver of successful coordination (Cui et al., 2012; Pan et al., 2017; Jiang et al., 2012), we found that neural coupling did not independently predict coordination after behavioral and dispositional factors were accounted for. Instead, its role emerged only in moderation analyses. Mechanistically, coupling within the premotor–parietal network is well suited to rhythmic interpersonal coordination, as premotor circuits support temporal prediction and sensorimotor timing, while inferior parietal regions contribute to mapping a partner's actions onto one's own action representations, enabling rapid mutual adjustment (Novembre et al., 2014; Penhune and Zatorre, 2019; Sebanz and Knoblich, 2009). From an interaction neuroscience accounts, interbrain synchrony is often viewed as an interaction-dependent index that tracks the degree of neural alignment required by coordination demands, rather than a uniform driver of performance (Czeszumski et al., 2020; Kelsen et al., 2022). Consistent with this view, motor and parietal systems are repeatedly implicated in flexible joint-action control, particularly when coordination requires ongoing monitoring and adaptive adjustments between partners (Bolt and Loehr, 2021). Neural synchrony enhanced performance when shared flexibility was low but offered little to no benefit when synergy was strong. This selective compensation pattern is consistent with predictive-processing accounts, which emphasize that coordination relies on reducing prediction error via shared or aligned internal models

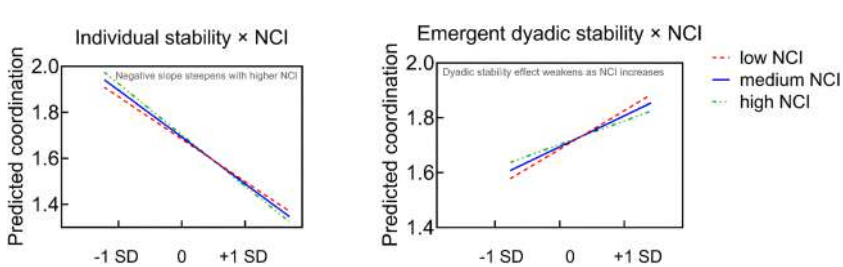

Fig. 6. Moderation effects of neural coupling (NCI) on behavioral stability components. Predicted coordination is shown as a function of each stability predictor across low, medium, and high NCI. Neural coupling strengthened the negative effect of overall stability (left, S_{mean}) and compensated for low behavioral synergy (right, S_{dyads}) by improving coordination when synergy was weak.

(Clark, 2013; Friston, 2010). When shared flexibility is low, increased interbrain synchrony may reflect enhanced neural alignment supporting shared temporal prediction during joint action (Sebanz and Knoblich, 2009). As shared flexibility increases, prediction error is reduced at the behavioral level, rendering additional neural alignment less necessary, consistent with the view that hyperscanning measures primarily track interactional demands rather than serving as uniform performance drivers (Czeszumski et al., 2020).This pattern is consistent with views that interbrain synchrony reflects compensatory engagement or heightened monitoring under increased interactional difficulty (Fishburn et al., 2018; Wang et al., 2020). When partners fail to jointly relax their control tendencies and cannot form a stable dyadic attractor, the system may rely on increased neural alignment to maintain coordination. Conversely, when shared flexibility at the behavioral level has already produced a robust dyadic solution, neural coupling becomes redundant. The moderating effect of neural coupling, therefore, reflects its sensitivity to the dyad's behavioral dynamics rather than functioning as a stable or context-independent driver. Neural synchrony appears to be recruited when behavioral processes alone are insufficient to sustain coordination, consistent with a state-dependent compensatory role. Thus, neural synchrony does not act as a universal marker or cause of successful coordination but as a context-dependent support mechanism that emerges when the dyadic system struggles to self-organize. These findings reveal a multilevel control architecture in which shared flexibility forms the primary foundation of coordinated action, while interbrain synchrony provides selective assistance only when this foundation is unstable.

A further direction is to characterize the link between interbrain synchrony and shared flexibility using richer spatio-temporal representations of neural dynamics. The present work summarizes coupling with a channel-level synchrony index, which is interpretable but intentionally low-dimensional. Recent EEG and fNIRS studies have shown that decomposing periodic and aperiodic activity, constructing functional connectivity networks, and learning low-dimensional neural manifolds can provide sensitive markers of neural state and modulation (Yu et al., 2025).Applying these approaches to hyperscanning could enable network-level or manifold-level indices of interbrain alignment, and future work may examine whether such representations better track, predict, or mediate shared flexibility during dyadic coordination.

Beyond its theoretical contributions, the present work offers several practical implications for understanding and supporting coordinated action in applied settings. The finding that stable coordination depends on partners' willingness to relax rigid individual control rather than on individual precision suggests that training programs may benefit from emphasizing mutual adaptability, shared timing strategies, and flexible role negotiation. Such principles may be particularly relevant in domains requiring tight interpersonal coupling, including rehabilitation, sports performance, and collaborative skill learning. At the same time, several limitations should be acknowledged. The task involved a constrained rhythmic movement paradigm that may not fully capture the richness of naturalistic joint action, and the sample size restricted our ability to examine finer-grained individual differences or long-term interactional dynamics. Neural coupling was assessed through a targeted frequency band derived from wavelet coherence, which, although sensitive to compensatory processes, does not encompass the full spectrum of possible cross-brain interactions. Future studies will be necessary to extend these findings to more complex behavioral contexts and to integrate higher temporal resolution measures that can illuminate how neural and behavioral adjustments unfold over time.

Together, these findings advance a multilevel account of interpersonal coordination in which shared flexibility provides the primary foundation for stable joint action, dispositional structure shapes the conditions under which such synergy can emerge, and neural coupling offers selective support when behavioral alignment alone is insufficient. This hierarchical architecture highlights that coordinated behavior is neither solely an individual achievement nor a purely automatic emergent process. Instead, it reflects the continuous negotiation between partners as they balance internal control, interpersonal compatibility, and real-time adaptive mechanisms. By demonstrating how shared flexibility enables synergy and how neural processes are recruited when this behavioral foundation becomes unstable, the present study contributes a more integrated framework for understanding how two individuals form a functional collective system. Future work may extend this perspective by examining how these mechanisms unfold in naturalistic settings, how they generalize across social contexts and how targeted interventions might facilitate cooperative performance by enhancing the emergence of synergy.

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Ethics statement

The protocol was approved by the institutional ethics committee of Shanghai University of Sport in China (tracking number 102772019RT007).

Data and code availability

Data and code that support the current findings will be made available upon reasonable request for research purposes.

CRediT authorship contribution statement

Ruoyu Niu: Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Yanan Li:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Lei Liu:** Formal analysis, Data curation. **Yafeng Pan:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Ying Liu:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no competing interests.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.neuroimage.2026.121773.

References

Anderson, C., John, O.P., Keltner, D., Kring, A.M., 2001. Who attains social status? Effects of personality and physical attractiveness in social groups. J. Pers. Soc. Psychol. 81 (1), 116–132. https://doi.org/10.1037/0022-3514.81.1.116.

Bolt, N.K., Loehr, J.D., 2021. The motor-related brain activity that supports joint action: a review. Acta Psychol. 212, 103218. https://doi.org/10.1016/j.actpsy.2020.103218.

Campbell, J.D., Lavallee, L.F., 1993. Who am I? The role of self-concept confusion in understanding the behavior of people with low self-esteem. In: Baumeister, R.F. (Ed.), Self-Esteem: The Puzzle of Low Self-Regard. Springer US, pp. 3–20. https://doi.org/10.1007/978-1-4684-8956-9_1.

Chen, Y., Yu, Y., Niu, R., Liu, Y., 2018. Selective effects of postural control on spatial vs. Nonspatial working memory: a functional near-infrared spectral imaging study. Front. Hum. Neurosci. 12, 243. https://doi.org/10.3389/fnhum.2018.00243.

Clark, A., 2013. Whatever next? Predictive brains, situated agents, and the future of cognitive science. Behav. Brain Sci. 36 (3), 181–204. https://doi.org/10.1017/S0140525x12000477.

Cook, W.L., Kenny, D.A., 2005. The actor–Partner interdependence Model: a model of bidirectional effects in developmental studies. Int. J. Behav. Dev. 29 (2), 101–109. https://doi.org/10.1080/01650250444000405.

Cui, X., Bray, S., Reiss, A.L., 2010. Functional near infrared spectroscopy (NIRS) signal improvement based on negative correlation between oxygenated and deoxygenated hemoglobin dynamics. NeuroImage 49 (4), 3039–3046. https://doi.org/10.1016/j.neuroimage.2009.11.050.

Cui, X., Bryant, D.M., Reiss, A.L., 2012. NIRS-based hyperscanning reveals increased interpersonal coherence in superior frontal cortex during cooperation. NeuroImage 59 (3), 2430–2437. https://doi.org/10.1016/j.neuroimage.2011.09.003.

Czeszumski, A., Eustergerling, S., Lang, A., Menrath, D., Gerstenberger, M., Schuberth, S., Schreiber, F., Rendon, Z.Z., König, P., 2020. Hyperscanning: a valid method to study neural inter-brain underpinnings of social interaction. Front. Hum. Neurosci. 14, 39. https://doi.org/10.3389/fnhum.2020.00039.

Djalovski, A., Dumas, G., Kinreich, S., Feldman, R., 2021. Human attachments shape interbrain synchrony toward efficient performance of social goals. NeuroImage 226, 117600. https://doi.org/10.1016/j.neuroimage.2020.117600.

Fishburn, F.A., Murty, V.P., Hlutkowsky, C.O., MacGillivray, C.E., Bemis, L.M., Murphy, M.E., Huppert, T.J., Perlman, S.B., 2018. Putting our heads together: interpersonal neural synchronization as a biological mechanism for shared intentionality. Soc. Cogn. Affect Neurosci. 13 (8), 841–849. https://doi.org/10.1093/scan/nsy060.

Friston, K., 2010. The free-energy principle: a unified brain theory? Nat. Rev. Neurosci. 11 (2), 127–138. https://doi.org/10.1038/nrn2787.

Gray-Little, B., Williams, V.S.L., Hancock, T.D., 1997. An item response theory analysis of the Rosenberg self-esteem scale. Personal. Soc. Psychol. Bull. 23 (5), 443–451. https://doi.org/10.1177/0146167297235001.

Haken, H., Kelso, J.A.S., Bunz, H., 1985. A theoretical model of phase transitions in human hand movements. Biol. Cybern. 51 (5), 347–356. https://doi.org/10.1007/BF00336922.

Höffler, T.N., 2010. Spatial ability: its influence on learning with visualizations—a meta-analytic review. Educ. Psychol. Rev. 22 (3), 245–269. https://doi.org/10.1007/s10648-010-9126-7.

Hoshi, Y., 2007. Functional near-infrared spectroscopy: current status and future prospects. J. Biomed. Opt. 12 (6), 062106. https://doi.org/10.1117/1.2804911.

Huang, C., Dong, N., 2012. Factor structures of the Rosenberg Self-Esteem Scale: a meta-analysis of pattern matrices. Eur. J. Psychol. Assess. 28 (2), 132–138. https://doi.org/10.1027/1015-5759/a000101.

Huppert, T.J., Diamond, S.G., Franceschini, M.A., Boas, D.A., 2009. HomER: a review of time-series analysis methods for near-infrared spectroscopy of the brain. Appl. Opt. 48 (10), D280–D298. https://doi.org/10.1364/AO.48.00D280.

Jiang, C., Zhu, Y., Luo, Y., Tan, C.-S., Mastrotheodoros, S., Costa, P., Chen, L., Guo, L., Ma, H., Meng, R., 2023. Validation of the Chinese version of the Rosenberg self-esteem Scale: evidence from a three-wave longitudinal study. BMC Psychol. 11 (1), 345. https://doi.org/10.1186/s40359-023-01293-1.

Jiang, J., Dai, B., Peng, D., Zhu, C., Liu, L., Lu, C., 2012. Neural synchronization during face-to-face communication. J. Neurosci.: Off. J. Soc. Neurosci. 32 (45), 16064–16069. https://doi.org/10.1523/JNEUROSCI.2926-12.2012.

Kelsen, B.A., Sumich, A., Kasabov, N., Liang, S.H.Y., Wang, G.Y., 2022. What has social neuroscience learned from hyperscanning studies of spoken communication? A systematic review. Neurosci. Biobehav. Rev. 132, 1249–1262. https://doi.org/10.1016/j.neubiorev.2020.09.008.

Kelso, J.A.S., 1995. Dynamic patterns: The self-Organization of Brain and Behavior. The MIT Press pp. xvii, 334.

Kenny, D.A., Kashy, D.A., Cook, W.L., 2006. Dyadic Data Analysis. The Guilford Press pp. xix, 458.

Konvalinka, I., Roepstorff, A., 2012. The two-brain approach: how can mutually interacting brains teach us something about social interaction? Front. Hum. Neurosci. 6, 215. https://doi.org/10.3389/fnhum.2012.00215.

Konvalinka, I., Vuust, P., Roepstorff, A., Frith, C.D., 2010. Follow you, follow me: continuous mutual prediction and adaptation in joint tapping. Q. J. Exp. Psychol. (Hove) 63 (11), 2220–2230. https://doi.org/10.1080/17470218.2010.497843.

Kurihara, Y., Takahashi, T., Osu, R., 2022. The relationship between stability of interpersonal coordination and inter-brain EEG synchronization during anti-phase tapping. Sci. Rep. 12 (1), 6164. https://doi.org/10.1038/s41598-022-10049-7.

Leung, S.O., Wong, P.M., 2008. Validity and reliability of Chinese rosenberg self-esteem scale. New Horiz. Educ. 56, 62–69.

Lu, K., Xue, H., Nozawa, T., Hao, N., 2019. Cooperation makes a group be more creative. Cereb. Cortex 29 (8), 3457–3470. https://doi.org/10.1093/cercor/bhy215.

Niu, H., Li, X., Chen, Y., Ma, C., Zhang, J., Zhang, Z., 2013. Reduced frontal activation during a working memory task in mild cognitive impairment: a non-invasive near-infrared spectroscopy study. CNS Neurosci. Ther. 19 (2), 125–131. https://doi.org/10.1111/cns.12046.

Niu, R., Xu, X., Tang, W., Xiao, Y., Tang, R., 2024. Dance of two brains: interval subdivision in alternated condition enhances resistance to interference by others. NeuroImage 298, 120788. https://doi.org/10.1016/j.neuroimage.2024.120788.

Niu, R., Yu, Y., Li, Y., Liu, Y., 2019. Use of fNIRS to characterize the neural mechanism of inter-individual rhythmic movement coordination. Front. Physiol. 10, 781. https://doi.org/10.3389/fphys.2019.00781.

Novembre, G., Ticini, L.F., Schütz-Bosbach, S., Keller, P.E., 2014. Motor simulation and the coordination of self and other in real-time joint action. Soc. Cogn. Affect. Neurosci. 9 (8), 1062–1068. https://doi.org/10.1093/scan/nst086.

Pan, Y., Cheng, X., Zhang, Z., Li, X., Hu, Y., 2017. Cooperation in lovers: an fNIRS-based hyperscanning study: cooperation in Lovers. Hum. Brain. Mapp. 38 (2), 831–841. https://doi.org/10.1002/hbm.23421.

Peng, W., Zhou, Y., Chu, J., Liu, Z., Zheng, K., Yao, S., Yi, J., 2024. Factorial and criterion validities of the Chinese version of Rosenberg self-esteem scale among undergraduate students. Psychol. Res. Behav. Manag. 17, 4135–4144. https://doi.org/10.2147/PRBM.S494452.

Penhune, V.B., Zatorre, R.J., 2019. Rhythm and time in the premotor cortex. PLoS Biol. 17 (6), e3000293. https://doi.org/10.1371/journal.pbio.3000293.

Piper, S.K., Krueger, A., Koch, S.P., Mehnert, J., Habermehl, C., Steinbrink, J., Obrig, H., Schmitz, C.H., 2014. A wearable multi-channel fNIRS system for brain imaging in freely moving subjects. NeuroImage 85, 64–71. https://doi.org/10.1016/j.neuroimage.2013.06.062.

Repp, B.H., Su, Y.-H., 2013. Sensorimotor synchronization: a review of recent research (2006–2012). Psychon. Bull. Rev. 20 (3), 403–452. https://doi.org/10.3758/s13423-012-0371-2.

Rosenberg, M., 1965. Society and the adolescent self-image. Princeton University Press.

Schmidt, R.C., Richardson, M.J., 2008. Dynamics of interpersonal coordination. In: Fuchs, A., Jirsa, V.K. (Eds.), Coordination: Neural, Behavioral and Social Dynamics. Springer, pp. 281–308. https://doi.org/10.1007/978-3-540-74479-5_14.

Schmitt, D.P., Allik, J., 2005. Simultaneous administration of the Rosenberg Self-Esteem Scale in 53 nations: exploring the universal and culture-specific features of global self-esteem. J. Pers. Soc. Psychol. 89 (4), 623–642. https://doi.org/10.1037/0022-3514.89.4.623.

Sebanz, N., Knoblich, G., 2009. Prediction in joint action: what, when, and where. Top. Cogn. Sci. 1 (2), 353–367. https://doi.org/10.1111/j.1756-8765.2009.01024.x.

Tice, D.M., 1993. The social motivations of people with low self-esteem. Self-esteem: The puzzle of Low Self-Regard. Plenum Press, pp. 37–53. https://doi.org/10.1007/978-1-4684-8956-9_3.

Wang, Z., Wang, Y., Zhou, X., Yu, R., 2020. Interpersonal brain synchronization under bluffing in strategic games. Soc. Cogn. Affect. Neurosci. 15 (12), 1315–1324. https://doi.org/10.1093/scan/nsaa154.

Witkin, H.A., Moore, C.A., Goodenough, D.R., Cox, P.W., 1977. Field-dependent and field-independent cognitive styles and their educational implications. Rev. Educ. Res. 47 (1), 1–64. https://doi.org/10.2307/1169967.

Witkin, H.A., Oltman, P.K., Raskin, E., Karp, S., 1971. A Manual For the Embedded Figures Test. Consulting Psychologists Press.

Wu, Y., Zuo, B., Wen, F., Yan, L., 2017. Rosenberg self-esteem scale: method effects, factorial structure and scale invariance across migrant child and urban child populations in China. J. Pers. Assess. 99 (1), 83–93. https://doi.org/10.1080/00223891.2016.1217420.

Yu, H., Zeng, F., Liu, D., Wang, J., Liu, J., 2025. Neural manifold decoder for acupuncture stimulations with representation learning: an acupuncture-brain interface. IEEE J. Biomed. Health Inf. 29 (6), 4147–4160. https://doi.org/10.1109/JBHI.2025.3530922.

Zhang, X., Noah, J.A., Hirsch, J., 2016. Separation of the global and local components in functional near-infrared spectroscopy signals using principal component spatial filtering. Neurophotonics 3 (1), 015004. https://doi.org/10.1117/1.NPh.3.1.015004.

Zheng, L., Chen, C., Liu, W., Long, Y., Zhao, H., Bai, X., Zhang, Z., Han, Z., Liu, L., Guo, T., Chen, B., Ding, G., Lu, C., 2018. Enhancement of teaching outcome through neural prediction of the students' knowledge state. Hum. Brain Mapp. 39 (7), 3046–3057. https://doi.org/10.1002/hbm.24059.